



Talk to data

Phase 8 - Production readiness and controlled release - Annexes

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Table of Contents

1.	How to use this annex	5
2.	Activity 1: Confirm production scope and release boundary	6
2.1.	Production release boundary template	6
2.2.	Boundary-to-evidence mapping	6
2.3.	Suggested release boundary statuses	7
2.4.	Illustrative example: controlled production boundary	7
2.5.	Scope challenge questions	8
2.6.	Common boundary mistakes	8
2.7.	Evidence to retain	8
3.	Activity 2: Triage pilot findings and production-readiness backlog	9
3.1.	Production-readiness issue disposition register	9
3.2.	Issue disposition guide	9
3.3.	Priority assessment matrix	10
3.4.	Illustrative examples	10
3.5.	Pre-release hardening backlog template	11
3.6.	Release constraint log	11
3.7.	Residual-risk candidate template	12
3.8.	Triage meeting agenda	12
3.9.	Common triage mistakes	12
3.10.	Evidence to retain	13
4.	Activity 3: Complete pre-release hardening and regression checks	14
4.1.	Pre-release hardening backlog template	14
4.2.	Fix classification guide	14
4.3.	Regression check template	15
4.4.	Suggested non-regression coverage	15
4.5.	Illustrative examples	16
4.6.	Fix and regression capacity plan	16
4.7.	Behaviour-change record	17
4.8.	Hardening cut-off rule	17
4.9.	Common hardening mistakes	17
4.10.	Evidence to retain	18
5.	Activity 4: Confirm production control readiness	19
5.1.	Production control readiness checklist	19

5.2.	Production control review record.....	19
5.3.	Data, semantic and refresh readiness template	20
5.4.	Access lifecycle readiness template.....	20
5.5.	Audit and traceability evidence checklist	21
5.6.	Illustrative example: production control gap	21
5.7.	Control readiness status guide.....	21
5.8.	Common control-readiness mistakes.....	22
5.9.	Evidence to retain	22
6.	Activity 5: Confirm operational reliability and observability	23
6.1.	Operational readiness checklist	23
6.2.	Monitoring coverage matrix	23
6.3.	Operational threshold template	24
6.4.	Incident severity guide	24
6.5.	Alert routing template	25
6.6.	Pause and rollback decision guide	25
6.7.	Feedback-to-trace template	25
6.8.	Early-life support monitoring cadence	26
6.9.	Common observability mistakes	26
6.10.	Evidence to retain	26
7.	Activity 6: Prepare support, documentation, onboarding and controlled adoption readiness 27	
7.1.	Documentation readiness checklist	27
7.2.	Documentation set by audience.....	27
7.3.	User onboarding checklist	28
7.4.	Release notes template	28
7.5.	Known limitations template	28
7.6.	Support runbook outline	29
7.7.	Controlled adoption readiness plan.....	29
7.8.	Issue communication examples.....	29
7.9.	Documentation ownership plan	30
7.10.	Common documentation and onboarding mistakes.....	30
7.11.	Evidence to retain	30
8.	Activity 7: Consolidate operating model, budget and roadmap capacity	31
8.1.	Operating model capacity worksheet	31

8.2.	Run / fix / improve / expand roadmap model	31
8.3.	Cost and capacity estimate template	32
8.4.	Capacity reservation template	32
8.5.	Roadmap prioritisation guide	33
8.6.	Funding and approval record	33
8.7.	Illustrative example: capacity-aware roadmap	33
8.8.	Common operating-model and budget mistakes.....	34
8.9.	Evidence to retain	34
9.	Activity 8: Approve and execute controlled release, residual-risk acceptance and Phase 9 handover	35
9.1.	Controlled release decision record.....	35
9.2.	Final release approval checklist	35
9.3.	Residual-risk acceptance record	36
9.4.	Release package contents	36
9.5.	Release execution or handover plan	36
9.6.	Pause, rollback and narrowing authority	37
9.7.	Phase 9 handover template	37
9.8.	Post-release review criteria	37
9.9.	Expansion criteria	38
9.10.	Common release approval mistakes	38
9.11.	Evidence to retain	38

1. How to use this annex

These annexes provide optional working templates for **Phase 8: Production readiness and controlled release**. They are intended to help teams convert pilot evidence into a controlled production release decision, not to create a mandatory checklist or a parallel delivery methodology.

The main Phase 8 guide explains the release logic: what can go live, what must be fixed before release, what constraints apply, who owns the capability, what risks are accepted, what budget and support capacity are required, and how the capability should move into Phase 9 operation, adoption and improvement.

This annex provides reusable working material for that process. Teams should select only the templates that match the delivery intent, risk level, organisational maturity and production exposure of the release.

For a narrow controlled release, some templates may be lightweight. For a higher-risk release, broader user exposure, sensitive data, regulated use cases or executive decision support, the evidence, sign-off, support, monitoring, risk acceptance and cost material should be treated more formally.

The annexes are especially useful for:

- classifying issues from the pilot and deciding which must be fixed before release;
- defining the final production release boundary;
- tracking pre-release hardening, fixes and regression evidence;
- confirming production controls, monitoring and operational readiness;
- preparing support, documentation, onboarding and escalation material;
- sizing support, bug-fix, improvement and roadmap capacity;
- recording release approval, residual-risk acceptance and Phase 9 handover.

The templates should be adapted, not copied mechanically. A team should be able to explain why each selected artefact is needed for the release decision. If a template does not help decide, support, control or operate the production release, it should be simplified or skipped.

The central test for Phase 8 documentation is practical: can users understand what they can rely on, can support teams diagnose and respond to issues, can owners fund and govern the capability, and can decision-makers defend the release boundary and residual risks.

2. Activity 1: Confirm production scope and release boundary

2.1. Production release boundary template

Area	Production release decision
Release objective	What the first controlled production release is intended to enable.
Target users	Named user groups included in release.
Excluded users	User groups not included and why.
Data domains	Approved domains included in production scope.
Data assets	Production data assets, views, marts, APIs or semantic assets in scope.
Metrics	Approved metrics available for production use.
Dimensions	Approved dimensions, hierarchies and filters.
Question types	Supported question patterns.
Answer types	Number, table, chart, narrative, caveated answer, clarification or refusal.
Follow-up behaviour	Which follow-ups are supported and which are refused or clarified.
Explicit exclusions	Questions, domains, users, outputs or integrations excluded from release.
Known limitations	Limitations users and support teams must understand.
Evidence basis	Phase 7 evidence supporting inclusion.
Release constraints	Conditions under which the release is allowed.
Cost / capacity implication	Expected support, usage, monitoring or change impact.
Owner	Person or group accountable for the release boundary.
Approval status	Draft / reviewed / approved / rejected.

2.2. Boundary-to-evidence mapping

Release item	Included?	Phase 7 evidence	Production concern	Decision
Regional sales managers	Yes	High usage, repeated workflow fit, few support issues	Needs onboarding and clear limitation guidance	Include
Revenue by region	Yes	Frequently used, trusted, low issue rate	Current-month data caveat required	Include with caveat
Margin by product	No	High demand but inconsistent interpretation	Metric definition not production-ready	Exclude / roadmap
Customer-level drill-down	Limited	Useful but raised access concerns	Requires stricter access checks	Include for analysts only
Forecasting questions	No	Users requested it, but not tested	Unsupported answer type and high interpretation risk	Exclude

Executive summary answers	Limited	Valued by pilot users	Risk of overstatement	Include only with source and caveats
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2.3. Suggested release boundary statuses

Status	Meaning
Include	Supported by evidence and ready for controlled production release.
Include with caveat	Usable in production only with visible limitation or usage condition.
Include with constraint	Usable only for defined users, domains, volumes or support conditions.
Fix before inclusion	Valuable but requires Phase 8 hardening before release.
Monitor after release	Acceptable only with threshold, owner and response route.
Exclude from release	Not safe or mature enough for first production release.
Move to Phase 9 roadmap	Useful future candidate, but not needed for first release.
Return to earlier phase	Requires renewed data, semantic, architecture, validation or pilot work.

2.4. Illustrative example: controlled production boundary

Example — illustrative

Area	Decision
Release objective	Enable regional sales managers to ask governed questions on revenue performance.
Target users	20 regional sales managers and 3 sales operations analysts.
Excluded users	Finance, executives, account executives and external users.
Data domains	Sales revenue and pipeline summary.
Data assets	sales_revenue_mart, region_dimension, pipeline_summary_view.
Metrics	Net revenue, customer count, pipeline conversion.
Excluded metrics	Margin, profitability, forecast revenue, salesperson performance.
Question types	KPI lookup, trend, comparison, variance by region or segment.
Answer types	Number, table, short narrative, caveated answer, clarification and refusal.
Follow-up behaviour	Follow-ups allowed within same domain and approved dimensions.
Known limitations	Current-month revenue is provisional; pipeline refresh is daily, not real time.
Release constraints	No customer-level profitability, no forecasting, no individual performance analysis.
Cost / capacity implication	Expected 100–150 questions per week; daily support review for first two weeks.
Owner	Sales operations product owner.

2.5. Scope challenge questions

Use these questions before approving the release boundary:

- Is each included question type supported by Phase 7 evidence?
- Is the release scope smaller than or equal to the evidence-proven pilot boundary?
- Are any items included mainly because users requested them?
- Are all included metrics approved, owned and explainable?
- Are all included joins, filters, caveats and dimensions safe for production use?
- Can support teams explain what is in and out of scope?
- Can users understand the limitations without reading a technical document?
- Would the release still be safe if usage doubled?
- Does the proposed scope create additional cost, monitoring or support needs?
- Is there a named owner for every included domain, metric and release constraint?

2.6. Common boundary mistakes

Mistake	Why it matters
Including everything that worked once in pilot	One successful interaction is not production evidence.
Letting user demand define production scope	Demand proves interest, not readiness.
Treating caveats as a substitute for controls	Caveats do not fix weak semantics, unsafe joins or missing ownership.
Expanding to senior users too early	Decision impact and reputational risk increase sharply.
Including unsupported follow-ups	Multi-turn behaviour can silently expand scope.
Forgetting support impact	A broader boundary creates more incidents, questions and interpretation support.
Ignoring cost impact	More users and broader question types can increase model, query, monitoring and support cost.

2.7. Evidence to retain

- Approved production release boundary.
- In-scope and out-of-scope list.
- Boundary-to-evidence mapping.
- Known limitations and release constraints.
- Data, semantic and access owners for included scope.
- Cost and capacity assumptions.
- Decision record approving or rejecting the proposed boundary.

3. Activity 2: Triage pilot findings and production-readiness backlog

3.1. Production-readiness issue disposition register

Item	Description
Issue ID	Unique reference for tracking.
Source	Pilot feedback, support ticket, monitoring alert, hot fix, access issue, cost finding, governance review, user interview.
Description	Clear description of the finding or issue.
Affected user group	Users impacted by the issue.
Affected workflow	Business workflow or decision context impacted.
Affected component	Data, semantics, access, prompt, model, query generation, UI, monitoring, support, documentation, cost, governance.
Frequency	One-off, occasional, repeated, frequent.
Severity	Low / medium / high / critical.
Visibility	Low-risk internal, business user, senior stakeholder, executive, regulatory / audit-facing.
Decision impact	None, low, moderate, high, material decision impact.
Current status	Open, in review, fixed, accepted, deferred, rejected.
Proposed disposition	Blocker, pre-release fix, release constraint, accepted residual risk, monitored risk, Phase 9 backlog, enhancement, redesign / stop signal.
Required action	Fix, constrain, monitor, accept, document, defer, return to earlier phase, stop.
Owner	Named owner accountable for resolution or acceptance.
Cost / capacity impact	Expected effort, people or budget implication.
Release implication	Can release proceed? Under what condition?
Evidence required	Test, regression result, sign-off, monitoring rule, documentation, risk acceptance.
Review date	Date for decision or follow-up.

3.2. Issue disposition guide

Disposition	Use when	Release implication
Production blocker	The issue makes production unsafe, ungoverned or unsupportable.	Fix before release, narrow, redesign or stop.
Pre-release fix	The issue is fixable and required before controlled release.	Add to Phase 8 hardening backlog.
Release constraint	The issue can be managed by limiting users, scope, question types, volume or support conditions.	Release only with explicit constraint.
Accepted residual risk	The issue is known, understood and accepted by the accountable owner.	Record rationale, owner and review condition.

Monitored risk	The issue is acceptable only if actively watched.	Add monitoring threshold, owner and response route.
Phase 9 backlog	The issue is not required for first release but should be improved after go-live.	Add to run / fix / improve backlog.
Later enhancement	The item is valuable but outside release readiness.	Add to roadmap, not release decision.
Redesign / stop signal	The issue exposes structural weakness in data, semantics, architecture, controls or operating model.	Return to earlier phase, redesign, pause or stop.

3.3. Priority assessment matrix

Issue priority should not be based on technical severity alone.

Dimension	Low	Medium	High
User impact	Few users, workaround available.	Multiple users or repeated friction.	Critical user group or no practical workaround.
Decision impact	Informational only.	Supports operational or tactical decision.	Supports senior, financial, regulatory or high-visibility decision.
Trust impact	Annoying but explainable.	Could create hesitation or confusion.	Could damage confidence in the whole release.
Access / security impact	No exposure risk.	Permission ambiguity or unclear audit trail.	Data exposure, bypass risk or unauthorised access.
Support impact	Easy to explain or resolve.	Requires specialist support.	Likely to create repeated incidents or escalation.
Cost impact	Negligible.	Noticeable but manageable.	Could exceed budget, monitoring or support capacity.
Visibility	Internal delivery team.	Business users.	Executive, audit, regulator or external stakeholder.

3.4. Illustrative examples

Issue	Technical severity	Production priority	Reason	Disposition
Occasional formatting issue in analyst table export	Low	Low	Does not affect interpretation or decision quality.	Phase 9 backlog
Wrong caveat missing from executive revenue answer	Medium	High	Could lead senior users to over-trust provisional numbers.	Pre-release fix
Popular request for margin analysis	Not a defect	Medium	Strong demand, but metric not production-ready.	Exclude / roadmap

Access removal takes three days	Medium	High	Weak access lifecycle for production users.	Pre-release fix or release constraint
High-cost query triggered by broad follow-up	Medium	High	Could create cost spikes and performance issues.	Fix / monitor / constrain
Minor typo in onboarding guide	Low	Low	Does not affect safe use.	Fix if easy, otherwise Phase 9 backlog
Users repeatedly ask unsupported forecasting questions	Not a defect	Medium	Scope misunderstanding may create trust risk.	Onboarding update and refusal rule
SQL generation fails for a supported metric	High	High	Supported production question cannot be relied on.	Production blocker

3.5. Pre-release hardening backlog template

Fix ID	Issue linked	Required fix	Owner	Priority	Effort	Regression required	Release dependency	Status
FIX - 001	ISS-014	Add missing caveat to current-month revenue answer	Product / engineering	High	Small	Answer-format regression	Required before release	Open
FIX - 002	ISS-021	Tighten follow-up query cost guardrail	Engineering	High	Medium	Cost and query regression	Required before release	Open
FIX - 003	ISS-032	Update onboarding to clarify unsupported forecasting	Product / adoption	Medium	Small	Documentation review	Required before onboarding	Open

3.6. Release constraint log

Constraint	Reason	Applies to	Owner	User communication needed?	Monitoring needed?	Review trigger
No forecasting questions	Not validated and high	All users	Product owner	Yes	Unsupported-question rate	Phase 9 roadmap review

	interpretation risk					
Customer drill-down only for analysts	Access and interpretation risk	Sales operations analysts	Data owner	Yes	Access logs	After access review
Usage cap of 150 questions / week	Cost and support capacity	First release cohort	Platform owner	Yes	Cost dashboard	After four weeks

3.7. Residual-risk candidate template

Risk	Why not fixed before release?	Release condition	Accountable owner	Monitoring / control	Review date
Current-month revenue may be provisional	Source refresh timing cannot be changed before release	Caveat must be displayed in every relevant answer	Finance data owner	Caveat presence check	Monthly
Some unsupported questions likely during first week	Users are still learning scope	Refusal wording and onboarding material updated	Product owner	Unsupported-question dashboard	Two weeks after release

3.8. Triage meeting agenda

- Confirm release boundary under consideration.
- Review new and open Phase 7 findings.
- Classify each material issue by disposition.
- Identify production blockers and pre-release fixes.
- Confirm release constraints and monitored risks.
- Separate Phase 9 backlog items from later enhancements.
- Review cost, support and capacity implications.
- Confirm owners, evidence required and decision dates.
- Escalate structural blockers or unresolved risk acceptances.
- Update the release recommendation.

3.9. Common triage mistakes

Mistake	Why it matters
Treating all bugs by technical severity	Business impact depends on user, workflow, decision and visibility.
Hiding issues in a generic backlog	Release decisions require disposition, ownership and conditions.

Accepting fixes without regression evidence	A fix can change model behaviour, prompts, SQL or answer interpretation.
Treating user requests as release requirements	Demand proves interest, not readiness.
Deferring support issues to Phase 9	Support gaps become production incidents once users rely on the system.
Ignoring executive visibility	A small defect in a senior workflow can damage trust more than a larger issue in a low-risk workflow.
Assuming all residual risks are equal	Residual risk must be accepted by the right owner, not just noted by the project team.

3.10. Evidence to retain

- Production-readiness issue disposition register.
- Pre-release hardening backlog.
- Release constraint log.
- Residual-risk candidate list.
- Monitored-risk list with thresholds and owners.
- Phase 9 backlog and roadmap candidates.
- Triage decision record.
- Evidence requirements for each release-critical fix.

4. Activity 3: Complete pre-release hardening and regression checks

4.1. Pre-release hardening backlog template

Item	Description
Fix ID	Unique reference for the fix.
Linked issue	Issue ID from the production-readiness issue disposition register.
Fix description	What needs to change.
Fix type	Bug, control gap, documentation gap, onboarding gap, support gap, monitoring gap, cost issue, access issue.
Affected component	Data, semantics, metadata, prompt, model, retrieval, SQL generation, validation, UI, access, monitoring, support, documentation.
Release dependency	Required before release / required before onboarding / required before wider release / optional.
Owner	Named person or team accountable for the fix.
Support required	Engineering, data owner, semantic owner, product, support, security, finance, other.
Effort	Small / medium / large.
Risk if unfixed	Impact on release if the fix is not completed.
Regression required	Test or check required after the fix.
Evidence required	Test result, trace, screenshot, log, approval, updated documentation, monitoring rule.
Status	Open / in progress / fixed / regression passed / accepted / deferred / blocked.
Decision	Release, release with constraint, delay, narrow, return to earlier phase, stop.

4.2. Fix classification guide

Fix type	Examples	Typical evidence
Answer behaviour fix	Missing caveat, unclear refusal, overconfident wording, weak clarification.	Before / after answer examples, trace, answer-quality review.
Query or execution fix	Unsafe query pattern, slow query, missing limit, wrong aggregation.	SQL test, execution log, performance check, regression case.
Data / semantic fix	Metric caveat, incorrect join, missing dimension, stale metadata.	Data-owner sign-off, metric test, reconciliation, metadata update.
Access / security fix	Incorrect permission, slow revocation, weak masking, missing audit trail.	Access test, audit log, permission-change evidence.
Monitoring fix	Missing alert, weak threshold, incomplete trace, cost blind spot.	Dashboard update, alert test, log sample, threshold approval.
Support fix	Missing runbook, unclear escalation, no owner for common issue.	Updated runbook, support acceptance, escalation test.

Documentation / onboarding fix	Known limitation unclear, unsupported questions not explained.	Updated guide, release note, onboarding review.
Cost fix	Expensive query pattern, uncontrolled model route, usage spike.	Cost estimate, guardrail test, budget impact note.

4.3. Regression check template

Item	Description
Regression ID	Unique reference.
Fix linked	Fix ID being tested.
Test objective	What the regression check is proving.
Affected flow	User question, access check, retrieval, SQL generation, answer generation, support flow, monitoring, cost control.
Test input	Question, user profile, data condition, access scenario or operational event tested.
Expected result	What should happen.
Actual result	What happened.
Pass / fail	Result of the regression check.
Evidence retained	Logs, trace, generated SQL, answer output, screenshot, monitoring event, approval.
Owner	Person accountable for the check.
Re-test trigger	What future change should trigger this test again.

4.4. Suggested non-regression coverage

Change made	Minimum regression checks
Prompt change	Supported questions, refusals, caveats, answer format, overstatement check.
Model change	Golden questions, latency, cost, answer quality, refusal behaviour.
Retrieval change	Metadata selection, metric matching, caveat retrieval, unsupported-question handling.
SQL generation change	Query correctness, join safety, aggregation, filters, limits, execution guardrails.
Data or semantic change	Metric reconciliation, dimensions, caveats, lineage, sample questions.
Access rule change	Allowed user, denied user, revoked user, row / column restriction, audit log.
Monitoring change	Alert trigger, alert routing, dashboard visibility, severity classification.
UI or onboarding change	Scope visibility, limitation visibility, feedback route, support route.
Cost control change	Usage threshold, query cap, model-routing cost, high-cost question path.

4.5. Illustrative examples

Fix	Why required before release	Regression check	Release decision
Add current-month caveat to revenue answers	Without the caveat, users may over-trust provisional numbers.	Run revenue questions across month-to-date, prior month and trend examples. Confirm caveat appears where required.	Required before release
Tighten follow-up limits for broad questions	Broad follow-ups may trigger expensive or slow queries.	Test broad follow-ups, unsupported filters and high-volume queries. Confirm refusal, clarification or cap.	Required before release
Update onboarding to explain unsupported forecasting	Users repeatedly asked for forecasts during pilot.	Review onboarding material and refusal wording. Check unsupported forecasting questions are refused consistently.	Required before onboarding
Improve table export formatting	Useful but does not affect answer trust or operational support.	Basic UI check.	Phase 9 backlog
Add dashboard link to every answer	Valuable but not required for first controlled release.	Not needed for release.	Later enhancement

4.6. Fix and regression capacity plan

Role / skill	Needed for	Estimated capacity	Timing	Reserved?	Risk if unavailable
Engineering	Prompt, model, orchestration, query and UI fixes	5 days	Week 1–2	Yes / No	Fixes delayed or untested.
Data owner	Metric, caveat and data-quality sign-off	2 days	Week 1	Yes / No	Release boundary cannot be approved.
Semantic owner	Definition, join and dimension checks	2 days	Week 1–2	Yes / No	Unsupported questions may slip into scope.
Security / access owner	Access lifecycle and audit checks	1–2 days	Week 2	Yes / No	Release blocked or constrained.
Product owner	Prioritisation, release notes, known limitations	3 days	Throughout	Yes / No	Fixes lack release decision context.
Support owner	Runbook and escalation validation	2 days	Week 2	Yes / No	Release not supportable.

4.7. Behaviour-change record

Change	User-visible?	Support-visible?	Governance-visible?	Documentation update needed?	Release note needed?
Refusal wording changed for forecasting questions	Yes	Yes	No	Yes	Yes
Revenue caveat added to current-month answers	Yes	Yes	Yes	Yes	Yes
Cost guardrail added for broad follow-ups	Sometimes	Yes	Yes	Yes	Yes
Internal log tagging improved	No	Yes	Yes	No	No
Access revocation process changed	No	Yes	Yes	Yes	Yes

4.8. Hardening cut-off rule

Before starting hardening, agree a cut-off rule to prevent Phase 8 becoming an open-ended improvement loop.

Suggested rule:

Category	Treatment after cut-off
Production blocker	Must be fixed, or release is delayed / narrowed / stopped.
Pre-release fix	Must be fixed unless formally reclassified.
Release constraint	Can remain only if reflected in release boundary and user guidance.
Monitored risk	Can remain only with threshold, owner and response route.
Phase 9 improvement	Deferred to Phase 9 backlog.
Later enhancement	Deferred to roadmap.

4.9. Common hardening mistakes

Mistake	Why it matters
Fixing without regression	The fix may break another supported question, control or user flow.
Treating prompt changes as low risk	Prompt changes can alter answer style, caveats, refusals, SQL and trust behaviour.
Reassigning the delivery team too early	Critical fixes may be delayed or handled by people without system context.
Adding enhancements during hardening	The release target keeps moving and regression scope expands.

Updating behaviour without release notes	Users and support teams cannot explain changed behaviour.
Deferring support fixes to Phase 9	Support gaps become live incidents after release.
Ignoring cost impact of fixes	A safer fix may increase latency, model usage, monitoring or support effort.

4.10. Evidence to retain

- Pre-release hardening backlog.
- Fix evidence and linked issue IDs.
- Regression test results.
- Behaviour-change and control-change record.
- Updated release notes and known limitations.
- Updated runbooks, onboarding material or support guidance.
- Capacity reservation record for fixes and regression checks.
- Remaining unresolved issue list with release implication.

5. Activity 4: Confirm production control readiness

5.1. Production control readiness checklist

Control area	Readiness questions	Evidence to retain
Data assets	Are all production data assets approved for the release boundary?	Approved asset list, owner sign-off, lineage reference.
Refresh	Are refresh frequency, latency and freshness expectations clear?	Refresh schedule, freshness threshold, failure response.
Metrics	Are included metrics approved, implemented and explainable?	Metric cards, reconciliation evidence, owner approval.
Dimensions	Are dimensions, hierarchies and filters production-ready?	Dimension register, hierarchy rules, filter rules.
Joins and grain	Are approved joins, grains and aggregation rules enforceable?	Join rules, grain checks, sample query evidence.
Caveats	Are mandatory caveats visible where needed?	Caveat matrix, answer examples, caveat test evidence.
Metadata	Is runtime metadata complete enough for supported questions?	Metadata coverage check, retrieval test evidence.
Access	Are provisioning, permission changes and revocation ready?	Access test evidence, access lifecycle process.
Security	Are masking, row / column controls and exposure limits enforced?	Security test evidence, policy mapping, logs.
Auditability	Can the organisation trace who asked what, which data was used and what answer was produced?	Trace logs, audit log sample, retention decision.
Compliance / privacy	Are required compliance, privacy and data-owner conditions reflected in the system?	Review record, sign-off where required, residual-risk note.

5.2. Production control review record

Area	Owner	Review required?	Sign-off required?	Status	Conditions / comments
Data owner	Sales analytics owner	Yes	Yes	Approved	Revenue mart approved for regional users.
Semantic owner	Finance BI lead	Yes	Yes	Conditional	Current-month caveat must be displayed.
Security	Security architect	Yes	Yes	Approved	Row-level access test passed.
Privacy	Privacy officer	Yes	No	Reviewed	No personal data in release boundary.
Compliance	Compliance representative	Yes	Conditional	Pending	Needs final release note review.

Audit	Internal audit	Optional	No	Not required	Audit log retained for review.
Platform owner	Data platform lead	Yes	Yes	Approved	Monitoring and access logs available.

5.3. Data, semantic and refresh readiness template

Item	Production decision
Data asset	Name of table, view, mart, API or semantic layer asset.
Release use	Which question types or user groups depend on it.
Owner	Data or semantic owner.
Approved for release?	Yes / no / conditional.
Refresh frequency	Real time, hourly, daily, weekly, monthly, manual.
Acceptable freshness threshold	Maximum tolerated delay before caveat, alert or pause.
Caveat required?	Yes / no.
Quality checks	Completeness, reconciliation, freshness, schema, volume, anomaly.
Access controls	User groups, row / column rules, masking, aggregation limits.
Metadata readiness	Complete / partial / insufficient.
Release implication	Include, include with caveat, constrain, fix before release, exclude.

5.4. Access lifecycle readiness template

Scenario	Expected behaviour	Evidence required	Owner	Status
New approved user added	User receives correct access only within release boundary.	Provisioning log and access test.	Support / IAM owner	
User changes role	Access changes to match new role.	Role-change test.	IAM owner	
User leaves pilot / organisation	Access removed within agreed timeframe.	Revocation log.	IAM owner	
User requests unsupported domain	Access denied or answer refused.	Refusal trace and access log.	Product / engineering	
User lacks row-level permission	Restricted data not returned.	Row-level access test.	Data platform owner	
Support investigates issue	Support can view required logs without overexposure.	Support access review.	Security / support owner	

5.5. Audit and traceability evidence checklist

A production release should retain enough evidence to support diagnosis, governance and review.

Evidence	Minimum expectation
User identity	Who asked the question.
Timestamp	When the interaction occurred.
Question	Original user question and relevant follow-up context.
Scope classification	Supported, clarified, refused or out of scope.
Retrieved metadata	Metrics, dimensions, caveats or examples used.
Generated query	SQL or tool call generated, where applicable.
Executed query	Final executed query, with validation status.
Data source	Data assets used.
Answer	Final answer shown to the user.
Caveats / limitations	Caveats displayed or missing.
Cost / latency	Runtime, model calls, query cost or cost proxy.
Access decision	Allowed, denied, constrained or refused.
Error / incident link	Link to support ticket or incident if relevant.

5.6. Illustrative example: production control gap

Finding	Release implication	Response
Revenue metric approved, but current-month refresh can lag by 24 hours.	Release possible with caveat and freshness monitoring.	Display current-month caveat, add freshness alert, review after first month.
Customer drill-down works technically but access review is incomplete.	Cannot release to all users.	Restrict to sales operations analysts until access review is complete.
Audit logs record the answer but not retrieved metadata.	Weak traceability for support and governance.	Fix before release or constrain use to lower-risk questions.
Metric owner has approved pilot use but not production use.	Ownership gap.	Obtain production approval or exclude metric from release.
Revocation process is manual and untested.	Production access lifecycle risk.	Test revocation before release and assign owner / SLA.

5.7. Control readiness status guide

Status	Meaning	Release implication
Ready	Control is approved, enforceable and owned.	Release can proceed for this area.
Ready with caveat	Control is acceptable with visible limitation or monitoring.	Release with condition.
Ready with constraint	Control works only for specific users, domains or usage levels.	Constrain release boundary.
Fix before release	Control gap is material but fixable.	Add to hardening backlog.
Accept residual risk	Control gap remains but is knowingly accepted by accountable owner.	Record risk acceptance and review date.

Not ready	Control gap makes release unsafe or ungoverned.	Exclude, narrow, return to earlier phase or stop.
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5.8. Common control-readiness mistakes

Mistake	Why it matters
Treating pilot access as production access	Pilot access is often manually managed and not lifecycle-ready.
Approving data without refresh rules	Users may rely on stale or provisional data without knowing it.
Assuming caveats solve control gaps	Caveats help interpretation; they do not replace access, quality or semantic controls.
Missing owner sign-off	Production controls need accountable owners after the project team leaves.
Logging too little context	Support cannot diagnose issues without query, metadata, answer and access traces.
Making controls manual without capacity	Manual controls fail unless owned, scheduled, monitored and funded.
Forgetting cost of controls	Monitoring, audit retention, access management and quality checks all create run cost.

5.9. Evidence to retain

- Production control readiness summary.
- Approved data, semantic, metadata and refresh scope.
- Access lifecycle test evidence.
- Security, privacy, compliance, audit and data-owner review record.
- Audit and traceability evidence sample.
- Production-control gap list with release implications.
- Control ownership and maintenance responsibilities.

6. Activity 5: Confirm operational reliability and observability

6.1. Operational readiness checklist

Area	Readiness questions	Evidence to retain
Usage	What user volume and question volume are expected for the release boundary?	Usage assumption, pilot usage baseline, volume estimate.
Concurrency	How many users or requests may run at the same time?	Concurrency assumption, load / stress check where relevant.
Latency	What response time is acceptable for supported question types?	Latency threshold, pilot latency evidence, timeout rule.
Availability	What level of service availability is expected for first release?	Service expectation, support hours, downtime process.
Cost	What cost level is expected and what would trigger review?	Cost estimate, cost threshold, budget owner.
Data freshness	What freshness level is required for each production data asset?	Freshness threshold, refresh log, alert rule.
Failures	What failure types need monitoring and escalation?	Error taxonomy, alert route, incident trigger.
Unsupported questions	How will repeated unsupported questions be detected?	Unsupported-question dashboard or review process.
User issues	How will feedback, complaints and support tickets be linked to traces?	Feedback-to-trace process.
Pause / rollback	Who can pause, narrow or roll back the release?	Pause / rollback decision route.

6.2. Monitoring coverage matrix

Component	What to monitor	Example signal
User interaction	Questions asked, user group, scope classification.	Supported / clarified / refused / failed.
Metadata retrieval	Retrieved metric, dimension, caveat or example.	Retrieval success, missing metadata, low-confidence match.
Model calls	Model route, latency, cost, errors.	Token use, failed call, unexpected model path.
Query generation	SQL generated, validation status, blocked queries.	Unsafe SQL blocked, validation failure.
Query execution	Runtime, scanned data, failures, timeout.	Slow query, high-cost query, execution error.
Answer generation	Caveat inclusion, answer format, refusal behaviour.	Missing caveat, empty result, overlong answer.
Access decision	Allowed, denied, constrained or revoked access.	Denied access, role mismatch, revoked-user attempt.
Cost	Model cost, query cost, monitoring cost, support cost proxy.	Daily cost, cost per interaction, threshold breach.
Support	Tickets, issue type, response time, repeated user problems.	Open incidents, repeated unsupported questions.

Data freshness	Last refresh, failed refresh, stale asset.	Freshness breach, missing source update.
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6.3. Operational threshold template

Metric / signal	Normal range	Warning threshold	Critical threshold	Owner	Response
Average response latency	< 10 seconds	> 20 seconds	> 45 seconds	Platform owner	Investigate slow path, narrow heavy queries if needed.
Failed interactions	< 2%	> 5%	> 10%	Product / engineering	Review traces, classify issue, escalate if repeated.
Unsupported questions	< 10%	> 20%	> 30%	Product owner	Update onboarding, refusal wording or roadmap.
Daily model / query cost	Within budget	80% of daily threshold	100% of daily threshold	Finance / platform owner	Review usage, apply cap, adjust routing.
Data freshness delay	Within SLA	1 missed refresh	2 missed refreshes	Data owner	Add caveat, notify users, pause affected domain if needed.
Access failures	None expected	Repeated denied-user attempts	Incorrect access granted	Security / IAM owner	Investigate immediately, consider pause.
Missing caveat rate	0	Any repeated issue	Any issue on high-risk metric	Product / data owner	Fix before continued use if material.

6.4. Incident severity guide

Severity	Example	Expected response
Critical	Unauthorised data exposure, incorrect access, materially wrong executive answer, uncontrolled cost spike.	Immediate escalation, consider pause / rollback, notify accountable owners.
High	Repeated wrong answers for supported questions, missing caveats on material metrics, major latency failures.	Same-day investigation, fix or constrain release.
Medium	Repeated unsupported questions, confusing answer format, intermittent failures, support burden higher than expected.	Triage, monitor, update guidance or backlog.
Low	Minor formatting issue, isolated unclear wording, low-impact UI problem.	Fix if easy or add to Phase 9 backlog.

6.5. Alert routing template

Alert	Trigger	First owner	Escalation owner	User communication needed?
Data freshness breach	Asset misses agreed freshness threshold.	Data owner	Product owner	Yes, if affected answers are still available.
Cost threshold breach	Usage exceeds warning or critical threshold.	Platform owner	Finance owner	Only if usage is capped or degraded.
Access anomaly	Denied, revoked or unexpected access pattern.	Security / IAM owner	Risk owner	Yes, if user access is affected.
High failure rate	Failed interactions exceed threshold.	Engineering owner	Product owner	Yes, if issue affects supported questions.
Missing caveat	Caveat not shown for required metric.	Product / data owner	Governance owner	Yes, if users may have relied on answer.
Repeated unsupported questions	Unsupported-question threshold exceeded.	Product owner	Adoption lead	Possibly; update onboarding or guidance.

6.6. Pause and rollback decision guide

Trigger	Possible response
Confirmed unauthorised data exposure	Pause affected capability immediately and escalate.
Repeated material wrong answers	Disable affected question type, domain or metric until fixed.
Cost spike beyond agreed threshold	Apply usage cap, narrow scope or switch routing if safe.
Data freshness breach on material domain	Add warning, disable affected answers or pause domain.
Support capacity exceeded	Narrow user group, slow onboarding or increase support cover.
Monitoring unavailable	Delay release or pause until minimum observability is restored.
Owner unavailable for critical response	Pause affected release component until ownership is restored.

6.7. Feedback-to-trace template

Item	Description
Feedback ID	Unique user feedback or support ticket reference.
User group	User or group raising the issue.
Original question	User question or interaction.
Trace ID	Link to logs / trace.
Generated query	Query or tool call, where applicable.
Data assets used	Source assets, metrics and dimensions.

Answer shown	Answer and caveats shown to the user.
Issue type	Wrong answer, unclear answer, unsupported question, latency, access, cost, UI, other.
Severity	Low / medium / high / critical.
Owner	Person or team accountable for response.
Response	Fix, explain, constrain, monitor, defer, escalate.
User communication	What was communicated back to the user.

6.8. Early-life support monitoring cadence

Period	Recommended cadence	Focus
First 48 hours	Daily or twice daily review.	Access issues, failures, user confusion, cost spikes, major trust issues.
First two weeks	Daily or every other day.	Repeated questions, unsupported scope, latency, support burden, onboarding gaps.
First month	Weekly.	Trends, cost, issue closure, roadmap candidates, support capacity.
After stabilisation	Monthly or governance cadence.	Adoption, quality, cost, risk, improvement and expansion readiness.

6.9. Common observability mistakes

Mistake	Why it matters
Monitoring only infrastructure	T2D failures often sit in semantics, retrieval, caveats, access or interpretation.
No named alert owner	Alerts without owners become noise.
Cost review after month-end only	Cost spikes can occur quickly with broad questions, expensive models or repeated retries.
Logs not linked to user feedback	Support cannot diagnose what happened.
No pause trigger	Teams hesitate during incidents because nobody knows when to stop the release.
Dashboards built for the delivery team only	Support, product and owners need operational views they can act on.
Missing unsupported-question tracking	Repeated unsupported demand is often the first signal that scope or onboarding is weak.

6.10. Evidence to retain

- Operational reliability and observability summary.
- Usage, latency, availability, freshness and cost thresholds.
- Monitoring dashboard or log evidence.
- Alert routing and owner matrix.
- Incident severity and escalation guide.
- Pause / rollback decision route.
- Feedback-to-trace evidence.
- Early-life support cadence.
- Operational gaps with release implications.

7. Activity 6: Prepare support, documentation, onboarding and controlled adoption readiness

7.1. Documentation readiness checklist

Documentation area	Readiness question	Evidence to retain
Release boundary	Does the documentation explain what is in and out of scope?	Final release boundary, user-facing scope summary.
Supported questions	Are supported question types clear and understandable?	Supported-question guide, examples.
Exclusions	Are unsupported domains, questions and answer types explicit?	Exclusion list, refusal examples.
Known limitations	Are limitations visible without requiring technical knowledge?	Known limitations page, onboarding material.
Responsible use	Do users understand what the system can and cannot be relied on for?	Responsible-use guide.
Behaviour changes	Are Phase 8 fixes and behaviour changes documented?	Release notes, behaviour-change record.
Support route	Do users know how to raise issues and where to get help?	Support route, escalation guide.
Runbooks	Can support teams diagnose common issues?	Support runbook, incident playbook.
Governance evidence	Can owners defend the release decision and residual risks?	Approval record, residual-risk log.
Versioning	Is documentation versioned and aligned to the production release?	Version history, owner record.
Maintenance	Is there an owner and cadence for keeping documentation current?	Documentation ownership plan.

7.2. Documentation set by audience

Audience	Documentation needed	Main purpose
Users	Quick-start guide, supported questions, known limitations, responsible-use guidance, issue-reporting route.	Help users understand what they can rely on.
Support teams	Runbook, triage guide, common issues, escalation path, pause / rollback communication.	Help support teams diagnose and respond.
Product / delivery owners	Release notes, backlog, roadmap, behaviour-change record, regression evidence.	Manage release, fixes and improvement.
Data / semantic owners	Approved metrics, caveats, refresh rules, data scope, semantic constraints.	Maintain trusted data and definitions.
Platform / engineering owners	Deployment notes, monitoring dashboards, alert routes, logs, technical dependencies.	Operate and maintain the service.

Governance / risk owners	Release boundary, residual-risk acceptance, access controls, audit evidence, sign-off record.	Defend the release decision and oversight model.
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7.3. User onboarding checklist

Area	What users should understand
Purpose	What the controlled production release is for.
Scope	Which questions, domains and answer types are supported.
Limitations	Which questions are not supported and why.
Trust	How to interpret answers, caveats, clarifications and refusals.
Evidence	Where sources, caveats or traces are visible, if exposed to users.
Responsible use	What decisions the system can support and where human review remains required.
Feedback	How to report wrong, unclear, slow or unexpected answers.
Support	Who to contact and expected response route.
Changes from pilot	What has changed since Phase 7.
Roadmap	What may come later and what is deliberately not included now.

7.4. Release notes template

Item	Description
Release version	Version or release identifier.
Release date	Planned or actual release date.
Release objective	What this release enables.
Users in scope	User groups included.
Data / domains in scope	Domains and assets included.
Supported questions	Question types supported.
Key exclusions	Questions, users, domains or features excluded.
Behaviour changes	Changes since pilot or previous release.
Fixed issues	Release-critical issues fixed in Phase 8.
Known limitations	Remaining limitations users should understand.
Release constraints	Conditions under which the release is allowed.
Support route	How users raise issues.
Owner	Product or release owner.

7.5. Known limitations template

Limitation	User impact	Required user behaviour	Owner	Review trigger
Current-month revenue may be provisional.	Answers may change after source refresh.	Treat current-month figures as indicative.	Finance data owner	Month-end close.
Forecasting questions are not supported.	System will refuse or redirect these questions.	Use existing forecasting process.	Product owner	Phase 9 roadmap review.
Customer drill-down limited to analysts.	Managers may not see customer-level details.	Ask sales operations for detailed analysis.	Data owner	Access review complete.

Pipeline data refreshed daily.	Same-day changes may not appear.	Use operational CRM for real-time updates.	Sales operations owner	Refresh improvement approved.
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7.6. Support runbook outline

Section	Content
Service overview	What the released capability does and who uses it.
Release boundary	Supported users, domains, questions and exclusions.
Common issues	Wrong answer, unclear answer, refusal, latency, access issue, missing caveat, unsupported question.
First checks	User, question, trace ID, data asset, metric, generated query, answer, caveats, timestamp.
Triage route	Product, data, semantic, engineering, access, support or governance owner.
Severity guide	Low / medium / high / critical with examples.
Escalation path	Who to contact and when.
Pause / rollback route	Conditions and decision owner.
User communication	How to respond to users during incidents.
Evidence retention	What must be logged and where.

7.7. Controlled adoption readiness plan

Item	Decision
First production cohort	Which users are onboarded first.
Onboarding format	Live session, recording, written guide, office hours, embedded support.
Onboarding owner	Person or team responsible.
Support model during first weeks	Office hours, dedicated channel, service desk, named support owner.
Feedback route	How users report issues or requests.
Adoption signals	Usage, repeat usage, unsupported questions, support tickets, satisfaction, trust feedback.
Review cadence	Daily / weekly / monthly review during early-life period.
Expansion condition	What must be true before more users are added.

7.8. Issue communication examples

Scenario	User-facing communication
Unsupported question	“This question is not supported in the current release. The release currently supports revenue and pipeline questions by region and segment, but not forecasting or margin analysis.”
Known data freshness caveat	“This answer uses daily refreshed pipeline data. Same-day CRM updates may not be reflected.”
Incident affecting answer quality	“We have identified an issue affecting answers for current-month revenue. Please do not rely on this answer type until we confirm resolution.”
Release constraint	“Customer-level drill-down is available only to sales operations analysts in this release because access controls are still being reviewed for wider users.”

Behaviour change after fix	“Current-month revenue answers now include a provisional-data caveat. This was added to avoid over-reliance before month-end close.”
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7.9. Documentation ownership plan

Document	Owner	Review cadence	Update trigger
User guide	Product owner	Monthly during early life	Scope change, user confusion, new limitation.
Known limitations	Product / data owner	Weekly during first month	New caveat, data issue, release constraint.
Support runbook	Support owner	Weekly during first month	New incident, escalation change, repeated issue.
Release notes	Release owner	Every release	Fix, behaviour change, scope change.
Responsible-use guide	Product / governance owner	Quarterly or on major change	Risk change, new user group, new answer type.
Monitoring / alert guide	Platform owner	Monthly	New alert, threshold change, incident learning.

7.10. Common documentation and onboarding mistakes

Mistake	Why it matters
Treating documentation as user guidance only	Support, governance, product and platform teams also need release-ready documentation.
Reusing pilot material unchanged	Pilot documents may not reflect final fixes, constraints or production ownership.
Hiding limitations in technical language	Users need plain-language boundaries to avoid over-reliance.
No support route in user material	Users will escalate informally or stop trusting the system.
No release notes for behaviour changes	Users and support teams cannot explain why answers changed.
No documentation owner	Documentation becomes stale as soon as Phase 9 begins.
Onboarding too broad	Users may assume the system supports more than the controlled release boundary.

7.11. Evidence to retain

- Production documentation readiness summary.
- User onboarding material.
- Release notes and behaviour-change record.
- Known limitations and responsible-use guidance.
- Support runbook and escalation route.
- Controlled adoption readiness plan.
- Documentation ownership and maintenance plan.
- Evidence that users and support teams were onboarded before access.

8. Activity 7: Consolidate operating model, budget and roadmap capacity

8.1. Operating model capacity worksheet

Area	Capacity question	Evidence / input
Run support	Who will handle day-to-day user issues, access questions and service checks?	Support model, pilot support findings, expected user volume.
Bug fixing	Who will fix production defects after release?	Phase 8 hardening backlog, unresolved issue list.
Data / semantic maintenance	Who will maintain metrics, caveats, metadata, refresh rules and data-quality issues?	Data owner sign-off, semantic scope, control gaps.
Platform / engineering	Who will maintain orchestration, model routing, deployment, logging and monitoring?	Architecture owner, operational readiness findings.
Monitoring and incident response	Who reviews alerts, investigates incidents and communicates impact?	Alert routing, incident process, early-life support plan.
Onboarding and adoption support	Who supports first users and captures early-life feedback?	Onboarding plan, controlled adoption plan.
Governance and risk review	Who reviews residual risks, constraints, changes and expansion requests?	Risk acceptance, governance cadence.
Roadmap delivery	Who delivers approved improvements and expansion?	Phase 9 roadmap, capacity plan.

8.2. Run / fix / improve / expand roadmap model

Lane	Purpose	Typical work	Capacity owner
Run	Keep the live release stable.	Support, access changes, monitoring review, incident response, cost review.	Support / platform / product.
Fix	Resolve production defects and known issues.	Bug fixes, regression checks, control fixes, documentation corrections.	Engineering / data / product.
Improve	Make the current release better without expanding scope materially.	Better onboarding, clearer caveats, better monitoring, answer-quality improvements.	Product / engineering / support.
Expand	Add new users, domains, questions, integrations or higher-risk use cases.	New data scope, new semantic work, new controls, new validation, broader rollout.	Product / data / platform / governance.

8.3. Cost and capacity estimate template

Cost / capacity area	Estimate	Owner	Funding route	Review cadence	Notes
Model usage		Platform / finance		Weekly during early life	Include expected and threshold usage.
Query / warehouse cost		Data platform		Weekly / monthly	Include high-cost query risk.
Infrastructure		Platform owner		Monthly	Hosting, orchestration, storage, networking.
Monitoring and logging		Platform owner		Monthly	Dashboards, alerting, log retention.
Support capacity		Support owner		Weekly during early life	First-line and second-line support.
Engineering fixes		Engineering owner		Weekly	Reserved capacity for early defects.
Data / semantic maintenance		Data / semantic owner		Weekly / monthly	Metric, caveat, metadata and refresh issues.
Onboarding and adoption support		Product / adoption lead		Weekly during early life	User guidance, office hours, feedback.
Governance / risk review		Governance owner		Monthly or decision-based	Residual risk, constraints, expansion approvals.
Roadmap delivery		Product owner		Monthly / quarterly	Improvements and expansion.

8.4. Capacity reservation template

Role / team	Required for	Minimum capacity	Duration	Reserved?	Risk if not reserved
Product owner	Release decisions, prioritisation, user feedback, roadmap			Yes / No	No clear trade-off owner.
Engineering	Fixes, regression, orchestration changes, monitoring fixes			Yes / No	Defects remain unresolved.
Data owner	Data issues, refresh exceptions, source changes			Yes / No	Data risks remain unowned.
Semantic owner	Metric, caveat, dimension and definition changes			Yes / No	Answer trust degrades.

Support owner	User support, incident triage, communication			Yes / No	Users escalate informally.
Platform owner	Monitoring, cost, reliability, deployment			Yes / No	Operational issues are not acted on.
Security / risk	Access issues, risk review, residual-risk decisions			Yes / No	Control issues delay or block release.

8.5. Roadmap prioritisation guide

Candidate item	Run / fix / improve / expand	Evidence basis	Capacity needed	Cost impact	Decision
Add missing caveat monitoring	Fix	Phase 8 control gap	Engineering + product	Low	Include before or soon after release.
Add finance users	Expand	User demand, not yet piloted	Data + semantic + support	Medium	Defer until readiness confirmed.
Improve unsupported-question guidance	Improve	Pilot confusion	Product / onboarding	Low	Include in early Phase 9.
Automate access revocation report	Improve / control	Manual access process	Platform / IAM	Medium	Prioritise if release cohort expands.
Add margin analysis	Expand	High demand but metric unresolved	Data + semantic + validation	High	Roadmap only after semantic readiness.

8.6. Funding and approval record

Item	Decision
Budget owner	Named owner accountable for run and change funding.
Approved run budget	Budget for operating the first release.
Approved change budget	Budget for bug fixes, improvements and controlled roadmap work.
Cost review cadence	Weekly / monthly / quarterly.
Cost escalation trigger	Threshold that requires review or intervention.
Capacity owner	Person accountable for securing required people.
Funding gaps	Unfunded areas that may constrain release or roadmap.
Release implication	Proceed, constrain, delay, narrow or stop.

8.7. Illustrative example: capacity-aware roadmap

Lane	Item	Timing	Capacity assumption	Release implication
Run	Daily monitoring review for first two weeks	Week 1–2	Product + platform 30 minutes / day	Required for release.

Fix	Resolve missing caveat edge cases	Week 1	Engineering 2 days + product review	Required before release.
Improve	Add better examples to onboarding guide	Week 2–3	Product 1 day	Can happen after release.
Expand	Add finance user group	Month 2+	Data, semantic, support and access review	Only after first release stabilises.
Expand	Add margin questions	Later roadmap	Metric definition and validation required	Return to semantic readiness first.

8.8. Common operating-model and budget mistakes

Mistake	Why it matters
Treating model cost as the whole cost	People, monitoring, support, testing, data refresh and governance often cost more than inference.
Assuming the delivery team will stay available	Teams often move on before hardening, early-life support and roadmap work are complete.
Funding release but not support	Users get access, but no one owns issues after go-live.
Mixing fixes and enhancements	Release-critical defects compete with nice-to-have roadmap items.
Expanding before stabilising	New users and domains multiply support, cost and trust risks.
Ignoring data and semantic capacity	T2D roadmap expansion usually depends on data and semantic work, not only engineering.
No cost trigger	Teams discover cost pressure only after the budget is already exceeded.

8.9. Evidence to retain

- Operating model and capacity estimate.
- Run-cost and change-budget view.
- Capacity reservation record.
- Run / fix / improve / expand roadmap.
- Funding and approval record.
- Cost and capacity gap log.
- Roadmap prioritisation record.
- Release implication of any unfunded capacity gaps.

9. Activity 8: Approve and execute controlled release, residual-risk acceptance and Phase 9 handover

9.1. Controlled release decision record

Item	Decision
Release decision	Release / release with constraints / further hardening / narrow / return / redesign / pause / stop.
Decision date	Date of decision.
Release version	Version or release identifier.
Release boundary	Users, domains, questions, answer types and usage limits approved.
Main rationale	Why this decision is appropriate.
Conditions	Conditions that must remain true for release to proceed.
Constraints	Scope, user, data, support, cost or monitoring constraints.
Release-critical fixes completed?	Yes / no / conditional.
Residual risks accepted?	Yes / no / pending.
Budget approved?	Yes / no / conditional.
Support accepted?	Yes / no / conditional.
Monitoring active?	Yes / no / conditional.
Onboarding complete?	Yes / no / scheduled.
Decision owner	Accountable owner approving release.
Review date	First post-release review date.

9.2. Final release approval checklist

Area	Release question	Status	Owner
Scope	Is the final release boundary approved and documented?		Product owner
Fixes	Are release-critical fixes complete and regression-tested?		Engineering / product
Controls	Are data, semantic, access, security and audit controls enforceable?		Data / security owner
Observability	Are monitoring, logging, alerting and thresholds active?		Platform owner
Support	Are runbooks, escalation routes and incident paths ready?		Support owner
Onboarding	Have first production users been onboarded?		Product / adoption lead
Cost	Are run cost, thresholds and budget owner confirmed?		Finance / platform owner
Residual risk	Have residual risks been accepted by accountable owners?		Risk / business owner
Change rights	Are change, expansion, pause and rollback rights clear?		Release owner
Phase 9 handover	Are owners, cadence, backlog and roadmap transferred?		Product / operating owner

9.3. Residual-risk acceptance record

Risk	Why it remains	Release condition	Accountable owner	Monitoring / control	Review trigger
Current-month revenue may be provisional.	Source refresh cannot be accelerated before release.	Caveat shown in all relevant answers.	Finance data owner	Caveat check and freshness monitoring.	Month-end close or caveat failure.
Unsupported questions expected in first week.	Users are still learning release boundary.	Refusal wording and onboarding updated.	Product owner	Unsupported-question dashboard.	Threshold breach or repeated confusion.
Manual access review during early life.	Automation not ready for first release.	Named owner and weekly review.	IAM / security owner	Access-change log.	Cohort expansion request.

9.4. Release package contents

Section	Content
Release decision	Decision, rationale, date, approvers and conditions.
Release boundary	Users, domains, questions, answer types, exclusions and constraints.
Fix evidence	Completed hardening items and regression evidence.
Control evidence	Data, semantic, access, security, audit and compliance evidence.
Observability evidence	Monitoring, alerts, logs, thresholds and incident route.
Support evidence	Runbooks, escalation paths, support owners and service expectations.
Onboarding evidence	User guidance, known limitations, responsible-use guidance and communication plan.
Cost evidence	Budget, cost thresholds, review cadence and owner.
Risk evidence	Residual risks, accepted risks, monitored risks and review triggers.
Phase 9 handover	Owners, cadence, backlog, roadmap and expansion criteria.

9.5. Release execution or handover plan

Step	Description	Owner	Status
Final release approval	Confirm decision and release conditions.	Release owner	
Production deployment or enablement	Deploy release or enable access for approved users.	Platform / engineering	
User access activation	Add approved users only.	IAM / support	
Monitoring activation	Confirm dashboards, alerts and logs are active.	Platform owner	
User communication	Send release note, scope, limitations and support route.	Product / adoption lead	
Support readiness check	Confirm support channel and escalation owners are live.	Support owner	

First-day review	Review usage, errors, access, cost and user issues.	Product / platform / support	
Early-life cadence	Confirm daily / weekly review rhythm.	Operating owner	

9.6. Pause, rollback and narrowing authority

Trigger	Possible action	Decision owner	Communication required
Unauthorised data exposure	Pause affected capability immediately.	Security / release owner	Yes
Repeated material wrong answers	Disable affected question type or domain.	Product / data owner	Yes
Cost threshold breach	Apply usage cap, model-routing change or temporary constraint.	Finance / platform owner	If user impact
Monitoring unavailable	Delay, pause or narrow release.	Platform / release owner	If service affected
Support capacity exceeded	Slow onboarding, narrow users or increase support cover.	Support / product owner	Yes
Data freshness breach	Add warning, disable affected domain or pause answers.	Data / product owner	Yes
Owner unavailable for critical issue	Pause affected release area.	Release owner	If user impact

9.7. Phase 9 handover template

Area	Handover decision
Live release boundary	What is live and what remains excluded.
Operating owner	Owner accountable for the live capability.
Product owner	Owner accountable for user value, backlog and roadmap.
Support owner	Owner accountable for user support and issue triage.
Platform owner	Owner accountable for service reliability, monitoring and cost.
Data / semantic owners	Owners accountable for data assets, metrics, caveats and refresh rules.
Risk / governance owner	Owner accountable for residual-risk review and change governance.
Review cadence	Daily / weekly / monthly operating rhythm.
Run backlog	Items needed to keep the service stable.
Fix backlog	Known defects and production issues to resolve.
Improve backlog	Improvements to current release experience or controls.
Expand backlog	Potential new users, domains, questions or integrations.
Expansion conditions	What must be true before scope grows.

9.8. Post-release review criteria

Area	Review question
Usage	Are the right users using the system at the expected level?
Trust	Are users relying on the answers appropriately?
Quality	Are supported questions answered correctly and consistently?

Scope	Are users repeatedly asking out-of-scope questions?
Support	Is support demand manageable?
Cost	Is usage within budget and cost thresholds?
Reliability	Are latency, failures and incidents within agreed thresholds?
Controls	Are access, audit, data and semantic controls holding in production?
Roadmap	Should Phase 9 focus on run, fix, improve or expand?

9.9. Expansion criteria

- Before adding users, domains, question types or integrations, confirm:
- the first production release is stable enough to absorb more usage;
- known production defects are not blocking trust or support;
- data and semantic readiness exists for the proposed expansion;
- access, security and audit controls can support the wider scope;
- monitoring and cost thresholds can handle additional volume;
- support and onboarding capacity is available;
- residual risks are reviewed and accepted by the right owners;
- the expansion is funded and prioritised against run, fix and improve work.

9.10. Common release approval mistakes

Mistake	Why it matters
Treating release as a ceremony	The decision must record scope, risks, owners and conditions.
Accepting residual risk informally	Risk acceptance needs accountable ownership, not project-team acknowledgement.
Releasing before support is live	Users will escalate informally or lose trust.
No pause authority	Teams hesitate during incidents because nobody can stop or narrow the release.
Handover without budget	Phase 9 starts with responsibility but no capacity.
Expanding immediately after go-live	Early demand can hide unresolved support, cost or control issues.
Losing the release boundary	Users and stakeholders start treating a controlled release as general availability.

9.11. Evidence to retain

- Controlled release decision record.
- Final release approval checklist.
- Residual-risk acceptance records.
- Final release package.
- Release execution or handover plan.
- Pause, rollback and narrowing authority.
- Phase 9 handover record.
- Post-release review criteria.
- Expansion criteria and first review date.